MIL-H-14506E
8 January 1970
SUPERSEDING
MIL-H-0014506D(MU
26 October 1967 and
MIL-H-14506B
28 June 1963

MILITARY SPECIFICATION

HOSE, AIR DUCT: FOR FILTER UNITS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a type of low pressure flexible hose fabricated in specific lengths.

1.2 Classification. The hose shall be made in the following lengths as specified (see 6.2):

Size 1 - 2 feet long
Size 2 - 6 feet long
Size 3 - 9 feet long
Size 4 - 4 feet long

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C5-19-916	-	Hose - Assembly and Detail.
DL136-45-45	-	Tester, Air Leakage, Hose Tube, Q49.
DL136-13-680	-	Tester, Crush, Q84.
DL136-41-3000	-	Apparatus, Static Diffusion, Q170.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UN I FORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing C5-19-916 shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

3.1.1.1 Outer covering. The outer covering shall be tubular stockinet made from nylon an shall meet the following physical requirements:

Width	····	as required
Denier	----------	210
Weight per square yard	----	4.4 ounces minimum

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3.1.2 Components. All components of the hose shall conform to the specifications and drawing listed on Drawing C5-19-916.

3.2 Manufacture. The hose shall be manufactured as specified on Drawing C5-19-916.

3.3 Physical properties.

3.3.1 Weight. The weight of the hose shall not exceed 5.0 ounces per foot of the original unstressed length when tested as specified in 4.4.4.1.

3.3.2 Resilience. The permanent set of the hose shall not exceed 2 percent of the original unstressed length when tested as specified in 4.4.4.2.

3.3.3 Extensibility The length of the hose shall increase at least 50 percent of the original unstressed length when tested as specified in 4.4.4.3.

3.3.4 Crush-resistance and permanent distortion. The hose shall maintain a minimum of 90 percent of its unstressed diameter when subjected to a 225 pound static load uniformly applied over a 4 inch length of the wire reinforced section of hose, when measured 2 minutes after application of the load. The hose shall return to within 98 percent of its unstressed diameter when measured within 2 minutes after the load is removed. The testing shall be in accordance with 4.4.4.4.

3.3.5 Flexibility. The hose shall be flexible when tested as specified in 4.4.4.5.

3.3.6 Low-temperature fracture resistance. The hose shall not fracture when tested as specified in 4.4.4.6.

3.3.7 Leakage. The hose shall not leak when subjected to an internal pressure of 5 + 1/2 pound per square inch gage (psig) when tested as specified in 4.4.4.7.

3.3.8 Freedom from odor. The hose shall be free from objectionable odor when tested as specified in 4.4.4.8.

3.3.9 Mustard resistance. The mustard-vapor resistance of the hose shall be at least 30 minutes when tested as specified in 4.4.4.9. Tests shall be conducted by Quality Assurance Laboratory, Edgewood Arsenal, Edgewood Arsenal, Maryland 21010.

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3.4 Preproduction. Prior to the start of regular production, a preproduction sample of hoses shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship The hose shall be free from damage such as worn or abraded stockinet and foreign matter such as dirt, oil, or viscous material inside or outside the hose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the tests in 4.4.4.9. Samples will be forwarded to Quality Assurance Laboratory, Edgewood Arsenal, Edgewood Arsenal, Maryland 21010.

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not 'been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Preprduction inspection.

4.3.1 Sample. A preproduction sample of 20 hoses of the size or sizes specified in the contract or order, shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production (see 6.2).

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4.3.2 Inspection procedure.

4.3.2.1 For examination. The entire sample of hoses shall be examined for all the requirements of the applicable drawing and this specification.

4.3.2.2 For nondestructive tests. Five hoses shall be taken at random from the preproduction sample and each tested in accordance with 4.4.4.1 through 4.4.4.8 inclusive.

4.3.2.3 For destructive tests. Five hoses shall be taken at random from the preproduction sample and tested for mustard resistance in accordance with 4.4.4.9.

4.3.4 Acceptance/rejection criteria. Failure of any hose to comply with the requirements of this specification when examined and tested as specified in 4.3.2 is reason for rejection of the preproduction sample.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the hoses produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105 Sampling for major defects 101 through 108 inclusive shall be in accordance with level S-4.

4.4.2.2 For destructive tests. Two sample hoses for the mustard resistance test shall be taken at random from the hoses that have passed the tests specified in 4.4.4.1 through 4.4.4.8 inclusive.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The hoses shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. Sample hoses for mustard resistance shall be tested in accordance with 4.4.4.9. The acceptance number shall be 0.

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4.4.3.3 Classification of defects.

(a) Hose (Drawing C5-19-916).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL	0.65 percent defective	
101		Weight	4.4.4.1
102		Resilience	4.4.4.2
103		Extensibility	4*4.4.3
104		Crush resistance and permanent distortion	4.4.4.4
105		Flexibility	4.4.4.5
106		Low-temperature fracture resistance	4.4.4.6
107		Leakage	4.4.4.7
108		Odor	4.4.4.8
	AQL	1.0 percent defective	
109		Component missing	
110		Component incorrectly assembled	
111		Inside diameter at ends	
112		Length of ends	
113		Overall length	
114		Workmanship (3.5)	
Minor:	AQL	2,5 percent defective	
201		Outside diameter of ends	
202		Maximum outside diameter	

4.4.4 Tests. Tests shall be conducted in the sequence listed:

4.4.4.1 Weight. The weight of the hose shall be as specified when the hose is weighed to the nearest ounce.

4.4.4.2 Resilience. The hose shall be suspended by one end, and a weight of 5 pounds shall be attached to the other end and permitted to hang freely. After 100 + 20 seconds, the weight shall be detached and the hose shall be left at rest for 100 ± 20 seconds. The length of the hose shall then be determined.

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4.4.4.3 Extensibility The hose shall be suspended by one end and a weight of 10 pounds shall be attached to the other end and allowed to hang freely. After 100 ± 20 seconds, the length of the hose shall be determined.

4.4.4.4 Crush-resistance and permanent distortion. The hose shall be tested for the requirements specified in 3.3.4 using a Q84 Crush Tester (Drawing DL 136-13-680) or a functionally equivalent crush testing apparatus.

4.4.4.5 Flexibility.

(a) Rotation. In a room maintained at 25° ± 5° C, the hose shall be placed horizontal, flat surface with 20 inches of its length overhanging the edge. The part of the hose on the flat surface shall be rotated through 360 degrees about its longitudinal axis, at a uniform rate of approximately 3 revolutions per minute (rpm). The hose shall be rotated a minimum of six revolutions. At no time during the rotation shall the *lower* half of the overhanging portion deviate from the vertical by more than 10 degrees.

(b) Coiling. In a room maintained at 25° ± 5° C, the hose shall be closely coiled around a rod 2 inches in diameter. There shall be neither any exposure of the wire in the hose nor any other damage. After release, there shall be no noticeable permanent set in the hose.

4.4.4.6 Low-temperature fracture resistance. The hose shall be subjected to a temperature of -55° + 5° C for 48 hours. Maintained at the aforementioned temperature, at least 20 inches of the hose shall be closely coiled around a rod 2 inches in diameter. The hose shall be allowed to return to room temperature and then tested for leakage as specified in 4.4.4.7. Failure of the leakage test shall be regarded as failing the low temperature requirement.

4.4.4.7 Leakage. One end of the hose shall be blanked off. The other end shall be connected to a source of compressed air through an appropriate cut-off valve. A pressure gage shall be placed in the line between the cut-off valve and the hose. All connections shall be tight. The hose shall be filled with air to 5 + 1/2 pound per square inch gage (psig) and the valve between the source of air and hose closed. The hose with the valve assembly shall then be disconnected from the source air. A period of two minutes shall be allowed for the air in the hose to reach equilibrium. There may be an initial pressure drop during this equalization period. A constant test gage pressure for a period of three minutes after equalization shall indicate the component under test does not leak. Leakage of the hose shall be determined with the Q49 Air Leakage Tester (Drawing DL136-45-45).

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4.4.4.8 Odor. The hose shall be free from objectionable odors when determined smelling.

4.4.4.9 Mustard resistance. The mustard-vapor resistance shall be determined with the Q170 Static Diffusion Apparatus (Drawing DL136-41-3000) as follows:

(a) Cut a 6 inch specimen from the hose, excluding soft ends; center and wrap a strip of indicator paper 4 inches in width completely around the exterior surface of the specimen.

(b) Distribute approximately 0.5 milliliter of liquid mustard along a 2 inch section of the interior surface near the center of the specimen.

(c) Plug both ends of the specimen with rubber stoppers.

(d) Determine mustard-vapor resistance.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The hoses shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 Packing, level C. The hoses, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When these items are procured for storage and issue as repair parts, preservation, packaging, packing and marking shall be as specified on the packaging data sheets which are identified by the Federal Stock numbers.

6. NOTES

6.1 Intended use. All four size hoses (see 1.2) are used with armored vehicle and hospital filter units. In addition, the size 1 (two foot) hose is used on the M14, M24, and M25 series masks.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The size of the hose required.
(c) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD-93 and ABCA-ARMY-STD-154. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - MU

User activities:

Navy - MC
Air Force - 84

International interest (see 6.3)

Preparing activity:

Army - MU(EA)

Project No. 4240-0303

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NO. (If applicable)

MATERIAL PROCURED UNDER A
DIRECT GOVERNMENT CONTRACT | SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES | NO (If "Yes", in what way?)

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*)

SUBMITTED BY (*Printed or typed name and activity – Optional*) | DATE

DD FORM 1426 (JAN 64) REPLACES EDITION OF 1 JAN 58 WHICH MAY BE USED.

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Commanding Officer
Edgewood Arsenal
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Edgewood Arsenal, Maryland 21010

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